Automated Growth Rate Analysis and Results Visualization of Tumor Spheroids

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Abstract— Tumor spheroids are widely recognized as robust 3D in vitro models that closely mimic the architecture and microenvironment of in vivo tumors, offering a more physiologically relevant platform for cancer research. However, traditional methods for spheroid analysis often rely on manual contouring and threshold adjustments, which are labor-intensive, prone to user bias, and unsuitable for high-throughput applications. To address these limitations, this project introduces an automated image analysis tool that transforms microscope images into quantitative measurements, eliminating the need for manual segmentation. The system leverages Otsu's thresholding method for automatic boundary detection and integrates regionProps calculations to extract key geometric properties such as area, volume, major and minor axes, and eccentricity. Furthermore, it incorporates an interactive visualization system featuring a histogram-based threshold slider, real-time segmentation feedback, and multi-layer canvas displays for enhanced clarity. For cases where automated segmentation fails, a manual mask mode with contour and brush tools allows for precise adjustments. To ensure measurement accuracy, the tool includes an automatic calibration mechanism using a known reference bar for real-world unit conversion. Data output is seamlessly exported as CSV files for further analysis. This streamlined and scalable approach significantly improves research efficiency, standardization, and reproducibility, providing an intuitive solution for high-throughput spheroid-based studies. By integrating automation with advanced visualization techniques, this project presents a novel method for precise and reproducible 3D cell culture analysis, contributing to advancements in cancer biology and therapeutic research.

1 Introduction

Tumor spheroids are widely used in cancer research as three-dimensional (3D) in vitro models that closely mimic the complex architecture and microenvironment of in vivo tumors, offering a more physiologically relevant alternative to traditional two-dimensional (2D) cell cultures [2] [6]. Unlike monolayer cultures, which poorly replicate the spatial organization and cell-cell interactions observed in tumors, 3D multicellular tumor spheroids (MCTS) exhibit critical features such as oxygen and nutrient gradients, extracellular matrix (ECM) interactions, and cellular heterogeneity that influence drug response and therapeutic resistance [5]. As a result, MCTS have emerged as a crucial platform for preclinical studies, allowing for the evaluation of drug efficacy and optimization of combinatorial radio(chemo)therapies under conditions that more accurately reflect the in vivo tumor response [1].

Accurate and reproducible measurement of spheroid size and volume over time is essential to evaluate the efficacy of treatment and understand tumor progression. Traditionally, these measurements are performed manually using software like ImageJ or proprietary microscopy tools, where users must manually contour the spheroid boundaries or adjust thresholds to segment the image. However, this approach is time consuming, prone to operator bias, and often inconsistent with different imaging conditions, making it unsuitable for high-throughput applications [3]. Although open source software such as CellProfiler and ImageJ has introduced basic segmentation routines, these methods often require batch by batch specific illumination corrections and threshold adjustments, limiting their robustness and scalability for large datasets [8]. Consequently, there remains a critical need for automated, flexible, and high-throughput image analysis tools that can reliably measure 3D tumor spheroids with minimal user intervention, facilitating more efficient and standardized data collection for drug screening and therapeutic research.

To address the limitations of manual spheroid measurement, our project aims to develop an automated analysis tool capable of accurately measuring the area and computing the volume of tumor spheroids from microscope images. Using image-processing techniques, the tool can eliminate the need for manual contouring, significantly reducing user variability and processing time. It incorporates an interactive visualization system that dynamically adjusts the segmentation threshold, allowing real-time exploration of spheroid structure. In addition, users can interact with a threshold slider to observe changes in spheroid boundaries, facilitating more intuitive selection of the optimal threshold for spheroid segmentation. For cases where automated segmentation fails validation, the tool provides a manual contouring mode, allowing users to draw custom masks for measurements to replace the failed cases. This dual-mode functionality ensures robust and reliable analysis, enhancing reproducibility and scalability for high-throughput applications.

An exist approach was demonstrated by Murphy et al. [4], who converted confocal microscopy images to TIFF files in ImageJ and processed them using custom MATLAB scripts. These scripts measured radius (converted into an equivalent spheroid area) and incorporated modifications to handle pimonidazole staining and blurred central regions caused by hypoxia and deoxygenation. Their methods underscore the feasibility of automating image processing workflows for more reliable and reproducible spheroid measurements, openly available code (Zenodo DOI: 10.5281/zenodo.5121093). Another approach was demonstrated by Xu's group [3], who used a software developed to automate the measurement of 3D tumor spheroids from bright-field microscopy images. They used an active contour algorithm (also known as "Snakes algorithm") to have a manual contour first and the algorithm refine the contour closer and closer to the actual spheroids' shape, which eventually causes the software being semi-automated. It also fixed scaling based on the image metadata, which cause an issue to process different scale of images.

Compared with the approached by Murphy et al. and Xu et al., which involves another software and initial manual image preparation, our method aims to fully automate the workflow by integrating advanced image processing directly into a single web-based tool, which is relatively user friendly, and there is no requirement to download additional software. While Murphy et al. primarily measured radius to convert it to an equivalent area and Xu et al. used Active Contour Model algorithm for segmentation, our solution directly employs Otsu thresholding with dynamic slider adjustment, allowing real-time refinement of the mask, which is particularly useful for visual feedback during segmentation. Besides, the automatic calibration is important to be able to process images in different scale. Our program will include

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anautomatic calibration function as well.

2 Background and Methods

2.1 Otsu's Thresholding Method

Otsu's method is an image processing technique used to automatically determine the optimal threshold to separate foreground objects from the background in grayscale images. This method calculates the threshold by maximizing the inter-class variance, which measures the spread between pixel intensity distributions of the foreground (spheroid) and background [7]. The primary advantage of Otsu's method is its ability to adaptively select the ideal threshold value without manual intervention, making it particularly suitable for high-throughput analysis of microscope images where lighting conditions and contrast may vary.

In our project, Otsu's method is applied to convert grayscale images of spheroids into binary masks. This process involves iterating through pixel intensity values to find the threshold that best separates the spheroid structure from its background noise. Once segmented, the binary mask is further processed to compute key geometric properties such as area, major and minor axis lengths, and eccentricity. This automatic segmentation eliminates the need for manual contouring, significantly reducing user variability and processing time.

Additionally, our program integrates Otsu's threshold value with a dynamic slider interface, allowing real-time adjustment of the segmentation threshold. This interactive component is coupled with a histogram visualization, generated using D3.js, that displays the distribution of pixel intensities across the image. Users can visually inspect the histogram to identify optimal segmentation points, enhancing the precision of boundary detection for each individual spheroid.

2.2 Region Properties Calculation (regionProps)

To extract meaningful geometric properties from the segmented spheroid masks, our tool leverages the computeRegionProps function. This function calculates:

- Area: The total pixel count of the segmented region.
- Major Axis Length: The longest diameter of the spheroid.
- Minor Axis Length: The shortest diameter perpendicular to the major axis.
- Eccentricity: A measure of the spheroid's deviation from a perfect circle.

These properties are derived by analyzing the pixel distribution and computing second moments to identify the main orientation and elongation of the spheroid structure. The calculations are performed automatically during the processing pipeline, and the results are displayed in a synchronized results table for easy interpretation.

2.3 Automatic Calibration

Our tool also includes an automatic calibration mechanism based on a red calibration bar of known size (0.5 mm) that is included in the captured images. During the processing step, the tool scans for this red bar and the pixel length of the bar is measured and used to compute the conversion ratio from pixels to millimeters. This calibration ensures that measurements of area and volume are consistent and accurately reflect the real dimensions, regardless of image magnification or pixel resolution.

2.4 Manual Mask Mode (Contour and Brush Tools)

Although Otsu's method and automatic segmentation are highly effective, there are cases where spheroids may be poorly segmented due to low contrast, noise, or overlapping boundaries. To address this, our program provides a Manual Mask Mode that includes:

1. Contour Tool: Allows users to manually trace the boundary of the spheroid.
2. Brush Tool: Provides a freehand drawing option for creating custom masks.

These manual adjustments enable researchers to correct segmentation errors or refine boundaries, ensuring accurate measurement even in challenging imaging conditions. After manual adjustments are made, the software recalculates geometric properties and updates the results table to reflect the modifications.

2.5 Histogram Visualization and Threshold Slider

Another core feature of our tool is the interactive histogram visualization, rendered using D3.js. The histogram represents the distribution of grayscale intensities across the spheroid image. The threshold slider is linked to this histogram, allowing users to dynamically adjust the cutoff point for segmentation. Changes made with the slider are reflected instantly on the binary mask, offering real-time feedback that helps users fine-tune the segmentation.

This interactive visualization enhances user control over the segmentation process, making it easier to adjust for variations in image quality, illumination, and contrast. It also serves as a diagnostic tool, allowing users to identify optimal segmentation thresholds visually.

3 Research

3.1 Data resource

The data for this project will be sourced from my current research on tumor spheroids, ensuring direct access to high-quality microscopy images. An example of the data is shown in Figure 1.

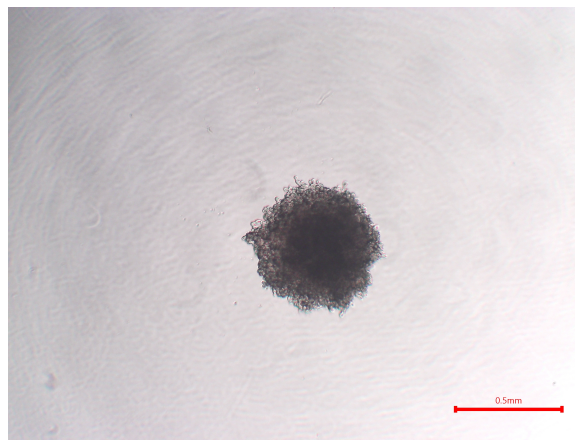

Fig. 1. An example of data source.

3.2 Evaluation

The project consists of two main components: (1) developing an automated analysis program to transform microscope images into digital data and (2) visualizing the segmentation and dynamically adjusting the cutoff point, offering real-time feedback that helps users fine-tune the segmentation.

To evaluate the effectiveness of this project during the semester, we conduct a series of tests comparing the automated spheroid analysis results to manually measured data. The primary evaluation metrics will include:

- Accuracy of Spheroid Detection and Measurement: the percentage of validated cases.
- Visualization Quality and Usability: The interactive visualization component will be tested for clarity, accuracy, and usability through feedback from lab members and other researchers.

Long-Term Evaluation and Future Implementation Beyond the course timeline, a comprehensive evaluation would involve direct comparison between spheroid area and volume calculated by this program against traditional manually contoured method by image J. Another more comprehensive evaluation is testing the this program on a larger dataset, including diverse spheroid types and imaging conditions. A

robust validation would require comparing results across different imaging techniques, such as brightfield, phase contrast, and fluorescence microscopy. In addition, I would explore the integration of machine learning-based segmentation to enhance detection accuracy, which would require training the algorithm on an expanded data set.

For long-term usability, the tool could be tested in external research labs to gather feedback from broader user groups. A potential future step would be integrating the tool with existing bioinformatics workflows or open-source platforms like Fiji (ImageJ) to facilitate adoption by the research community.

Currently, my lab provides access to necessary imaging data, computing resources, and research expertise to support the project. However, scaling up for machine learning integration or broader deployment would require additional funding and external collaborations. The evaluation plan set for this semester serves as a foundation for future enhancements, providing essential performance benchmarks and user feedback that will guide further development.

3.3 Technology

3.3.1 Real-time Histogram Visualization

In our tool, D3.js is used to generate an interactive histogram of grayscale pixel intensities. Each bar represents the count of pixels at a specific intensity level. The visualization updates in real-time when the threshold slider is adjusted, helping users identify optimal segmentation points, Figure 2.

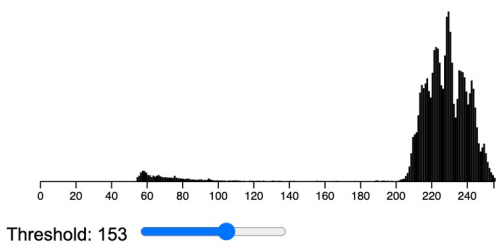

Fig. 2. Real-time Threshold Histogram

3.3.2 Interactive Threshold Slider

The Interactive Threshold Slider allows users to adjust the pixel intensity cutoff for segmentation directly. As the slider is moved, the tool recalculates the segmentation in real-time and updates the binary mask on the canvas. This live feedback mechanism, powered by JavaScript event listeners, enables users to fine-tune the segmentation threshold, optimizing boundary accuracy for various spheroid images, Figure 3.

3.3.3 Manual function: Multi-layer Canvas Display

Our tool employs a multi-layer canvas architecture to enhance image processing visualization. The Original Image Canvas presents the unprocessed spheroid, the Processed Mask Canvas overlays the binary segmentation, and the Manual Annotation Canvas allows for user-driven contouring. This architecture, managed by the HTML5 Canvas API, enables seamless toggling between automated and manual adjustments without disrupting the main view. It also provides a clear, interactive representation of segmentation changes as they occur as shown in Figure 4.

3.3.4 Color-based Calibration Visualization

Automatic calibration is achieved by detecting the red calibration bar present in each image. The tool identifies the bar based on its distinctive RGB values, measures its pixel length, and calculates the pixel-to-millimeter conversion ratio. This ratio is applied to all computed geometric properties, ensuring that measurements are accurate and consistent across varying magnifications.

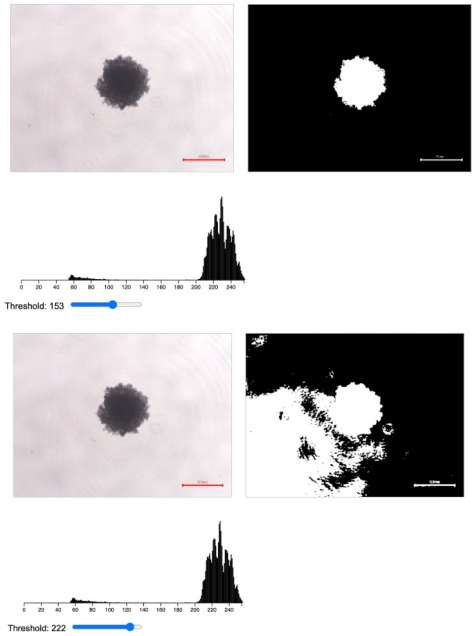

Fig. 3. Interactive Threshold Slider

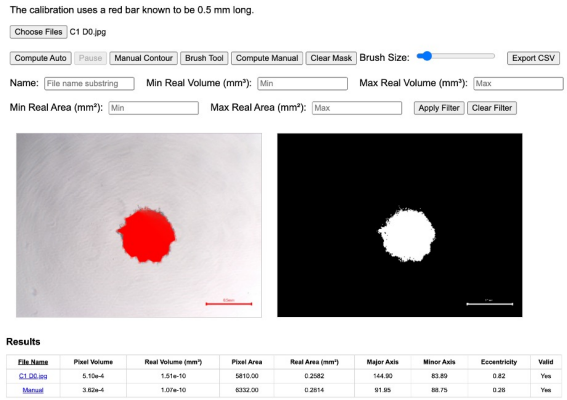

Fig. 4. Interactive Threshold Slider

3.3.5 Real-time Measurement Display

Our tool automatically calculates geometric properties of the segmented spheroid and displays them in a sortable HTML table. Among these, Eccentricity is particularly important for assessing how spherical or elongated the structure is. This information is important to assess whether the processed image is in the correct threshold range or not. As users adjust the segmentation threshold or apply manual contours, the measurements are recalculated instantly and reflected in the interface. This feature, powered by DOM manipulation, streamlines data inspection, ensuring users always have the latest results at their fingertips. The eccentricity is calculated by this formula, where a is the major axis length and b is the minor axis length.

$$Eccentricity = \sqrt{1 - \frac{b^2}{a^2}}$$

3.3.6 Data Table with Sorting and Filtering Capabilities

JavaScript Event Listeners was used to detect sorting and filtering actions. And HTML ¡table¿ Element was used to render data in a grid format, allowing sorting and filtering. The measurements are displayed in a dynamically sortable and filterable HTML table. Users can click

headers to sort by name, volume, area, axis length, or validity. Filtering allows users to refine the view based on custom criteria, simplifying the analysis of large datasets.

3.3.7 Exportable CSV for Data Analysis

JavaScript Blob API allows the creation of a binary large object (Blob) representing data, enabling the browser to treat it as a downloadable file. And the URL.createObjectURL Generates a URL for the Blob object that can be referenced for download. After segmentation and measurement are completed, our tool enables users to export the results in CSV format with a single click. The exported CSV includes all key measurements, such as pixel area, real area, pixel volume, real volume, and axis lengths, structured in columns for easy analysis. This allows for seamless integration with third-party software like Excel, MATLAB, and R, supporting further statistical analysis and reporting.

3.4 Results

3.4.1 First part:

Accuracy of Spheroid Detection and the percentage of validated cases. Among 3 trials of 144 images being processed by the program, only 5 percentage of images are marked as Not Valid. The Figure 5 is an example of spheroid that's marked as not valid.

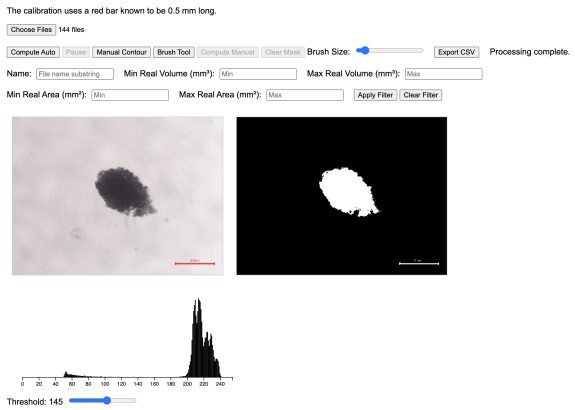

Fig. 5. Interactive Threshold Slider

3.4.2 Second part:

The interactive visualization component was tested for clarity, accuracy, and usability through feedback from lab members and other researchers. They reported that the interface is user friendly, all the fucntion can be accessed, but it fails on low-contrast cases images.

Operation Pipeline

1. Data and Scripts (Javascript and html) are uploaded in the same repository, called "Data and Script". When you run the script, please keep them in the same folder.

2. By double clicking the index.html file, you will have the interface ready to use.

3. Click choose file - select as many spheroids photo data as you want. Then click auto compute to have image data being processed and the results table display below. (you must click compute, otherwise, the photos are just loaded, they will not be processed.)

4. Check the eccentricity value and validation results. If the validation says "No", you can click the image name, and the binary masked images and it's threshold bar show up. Here you can double check whether the threshold been correctly chose.

Some additional function: 5. a). By clicking Export, you can export the data to a csv file. b). By tuning the filter, you can filter out the spheroids data that you want. For example, if you want only those sphroids that are bigger than 1.87e-10 Real Volume (mm³) and smaller than 3.96e-10 Real Volume (mm³), after filter, you will only have the data in between the area that you want. c). By clicking file name, you will have all data sorted by alphabetic sequence and number size. d). You can also use either manual contour or brush tool to analaze manually.

3.5 Timeline

I successfully completed the automated data analysis portion by April 1st and I aim to finalize the visualization component by April 20th.

Table 1. Project Milestones

Milestone	Description
Feb 25	Proposal
April 1	1st component
April 20	2nd component
April 25	Summary and finalize data
April 29	Presentation
May 6	Final report

4 Impacts

The automation of tumor spheroid analysis presents a transformative advancement in cancer biology research, offering significant improvements in efficiency, accuracy, and reproducibility. By eliminating the need for manual contouring and threshold adjustments, the developed tool mitigates user bias, reduces processing time, and streamlines data acquisition, enabling researchers to analyze larger datasets with enhanced consistency. This automated pipeline is not only optimized for spheroid analysis but also holds the potential for adaptation to other three-dimensional (3D) cell culture models, including tissue engineering constructs and organ-on-chip platforms. Such versatility broadens its impact, positioning it as a valuable tool for high-throughput screening and preclinical studies in regenerative medicine and drug testing.

Beyond its practical benefits, this project introduces an innovative integration of automation with interactive visualization techniques. The tool features real-time histogram-based threshold adjustment, layered canvas displays, and a manual contouring mode for fine-tuning segmentation. These capabilities allow researchers to intuitively interpret spheroid structure and boundary precision, ensuring that complex image datasets are processed with high accuracy and minimal human intervention. Additionally, the tool's web-based design enhances accessibility, making it a scalable resource for the biomedical community. This accessibility encourages standardized methodologies for spheroid-based studies, accelerating research in oncology, tissue engineering, and beyond.

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